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Heart rate recovery after aerobic and anaerobic tests: is there an influence of anaerobic speed reserve?

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ABSTRACT

The present study assessed if differences in the metabolic profile, inferred from the anaerobic speed reserve (ASR), would influence the dynamics of heart rate recovery (HRR) after two modes of exercise. Thirty-nine physical education students (14 females and 25 males) volunteered for this study. Participants carried out three separate testing sessions to assess maximal sprinting speed (MSS, 1st session), repeated sprint ability (RSA, 2nd session) and maximal aerobic speed (MAS) using the Université of Montreal Track Test (UMTT, 3rd session). ASR was defined as the difference between MSS and MAS. Heart rate was continuously registered throughout the tests and during the 5-min post-test recovery. To evaluate the influence of ASR on post-exercise, HRR comparisons between ASR-based groups [high ASR vs. low ASR] and sex groups (males vs. females) were performed. Significant differences ($P < 0.05$) were found between high ASR and low ASR groups of the same sex for indices of relative HRR after the RSA and UMTT. In addition, after the RSA test, males from the high ASR group had a significantly slower HRR kinetics compared with the males of the low ASR ($P < 0.05$) and the females of high ASR ($P < 0.05$); whereas females of the high ASR groups had a faster HRR kinetics compared with the females of low ASR group ($P < 0.05$). Our results showed that in males, post-exercise HRR could be related to the ASR, whereas in females, the influence of ASR is less clear.

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KEYWORDS

Heart rate; autonomic control; repeated sprint ability; metabolic profile; recovery

Introduction

Heart rate recovery (HRR) is the rate at which heart rate (HR) declines within the first minutes of post-exercise recovery. This response has been widely studied and it has been shown that its exponential feature can be attributed to the autonomic nervous system (ANS) and it is characterised by the parasympathetic reactivation and sympathetic withdrawal (Borresen & Lambert, 2008; Daanen, Lamberts, Kallen, Jin, & Van Meeteren, 2012). Moreover, the ANS is interlinked with other physiological systems (e.g., muscle metabolism) (Kiviniemi, Hautala, Kinnunen, & Tulppo, 2007) and therefore the evaluation of its responsiveness after an exercise load could provide an insight about the functional adaptation of the body, also finely responding to the recently applied training loads (Lamberts, Swart, Capostagno, Noakes, & Lambert, 2010). In this sense, the assessment of the autonomic function by means of non-invasive methods is an area of great interest both in clinical and sports settings.

HRR discriminates modes and intensities of exercise (i.e., endurance vs. resistance, maximal vs. submaximal exercises) (Daanen et al., 2012). It has been observed that HRR after maximal aerobic exercise is slower compared to submaximal exercise (Borresen & Lambert, 2008). Moreover, a slower HRR has been shown after repeated-sprint exercises in comparison with high-intensity interval training (HIIT; Buchheit, Laursen, & Ahmaidi,

2007). It appears that after supramaximal exercises, the sympathetic drive can last for prolonged time into the post-exercise recovery and contributes to the delayed parasympathetic reactivation (Al Haddad, Laursen, Ahmaidi, & Buchheit, 2009; Pierpont, Stolpmann, & Gornick, 2000), hence slowing down the HRR response when compared to submaximal exercises. Additionally, after supramaximal exercises (i.e., repeated-sprints), the blunted responsiveness of the HRR could be related to a higher anaerobic contribution during exercise and the elevation of adrenergic factors and local metabolites during recovery (Buchheit et al., 2007; Nakamura et al., 2009). Along this line of reasoning, those individuals showing distinct aerobic and anaerobic contributions to a given exercise load would therefore also display disparate post-exercise HRR responses.

The term anaerobic speed reserve (ASR) was first introduced by Blondel, Berthoin, Billat, and Lensel (2001) and refers to the difference between maximal speed and the velocity associated to the maximal oxygen uptake or maximal aerobic speed (MAS). It was put forth to test the hypothesis that running performance depends not only on MAS but also on the anaerobic distance capacity and the difference between maximal speed and critical velocity. Later, Bundle, Hoyt, and Weyand (2003) used the ASR as a means to predict all-out running speeds of any duration through direct measurements (i.e., oxygen deficit and oxygen consumption) of the maximum

speeds supported by the anaerobic and aerobic power of the runner. More recently, it has been proposed that ASR could be an important parameter that should be considered for training programming, especially for HIIT (Buchheit & Laursen, 2013). This view has been questioned (Boullosa, 2014; Boullosa & Abreu, 2014), as the influence of ASR on different modes of exercise has not been investigated and its effect on chronic adaptations were only marginally studied (Buchheit & Mendez-Villanueva, 2014). For instance, Buchheit, Hader, and Mendez-Villanueva (2012) found that ASR and ASR adjusted by MAS (i.e., ASR adjusted by the between-groups difference in MAS) had a low and a very high weight, respectively, in determining the numbers of HIIT sets maintained before exhaustion in a group of recreational team sports athletes. Also, in repeated-sprint exercise, Mendez-Villanueva, Hamer, and Bishop (2008) found a significant correlation ($r = 0.87$, $P < 0.05$) between the anaerobic power reserve (i.e., the difference between the maximal anaerobic power and the maximal aerobic power) and the fatigue index (i.e., the percentage decline from the first to the last sprint). However, it is not clear if those findings represent a mechanistic influence of the ASR or are the result of the chosen method of statistical analyses (Batterham & Hopkins, 2006). Nevertheless, it has been suggested that a lower use of the ASR could prevent an excessive peripheral physiological disturbance, thus sparing the anaerobic capacity and the neuromuscular function (Buchheit et al., 2012; Bundle et al., 2003). Accordingly, a lower ASR would induce less physiological disturbance, which could be reflected on the HRR response; yet to date there are no data regarding the possible influence of ASR on the post-exercise HRR. Consequently, if this is true, measuring HRR could be a useful tool not only for the assessment of the exercise load and to monitoring fatigue but also for establishing adequate recovery periods according to the metabolic profile of the individual.

Therefore, the purpose of this study was to assess if the differences in participant metabolic profile, as inferred from an index that integrates both MAS and maximal sprinting speed (MSS) (i.e., ASR), would influence the dynamics of HRR after two modes of exercise. Our hypothesis was that participants with a high ASR would exhibit a slower recovery compared to those with a low ASR.

Methods

Participants

Thirty-nine physical education students, 14 females (age: 23.6 ± 3.9 years, weight: 60.3 ± 7.7 kg; height: 159.7 ± 7.8 cm) and 25 males (age: 23.1 ± 3.1 years; weight: 74.5 ± 12.3 kg; height: 173.5 ± 8.8 cm) volunteered for this study. All participants were physically active and were engaged in different types of exercise. These included regular participation in sports practices (e.g., soccer, basketball, swimming, handball, etc.) and also resistance and endurance training. Prior to engaging in any test, participants were informed about the risks and benefits of the study and signed an informed consent form. The protocol of the study was approved by the local ethics committee and all methods are

in accordance with the Declaration of Helsinki for research on human subjects.

Procedures

To assess whether an individual's ASR could influence HRR after exercises of different metabolic demands, participants were required to participate in three separate testing sessions. During the first testing session, participants carried out a maximal sprint running test (3×40 m-all out sprints; 5 min of passive recovery); during the second session, a repeated sprint ability (RSA) test and during the third session, the Université of Montreal Track Test (UMTT). The testing sessions were separated by at least 72 h to avoid the effects of residual fatigue on performance. During the 2nd and 3rd testing sessions, HR was monitored to assess HRR after the RSA and UMTT tests. Participants were instructed to perform no strenuous exercise and to rest the day before the testing, and to abstain from alcohol or caffeine consumption in the 24 h prior to tests. All tests were performed between 07:00 and 11:00 hours, at approximately the same time of day for all individuals. The ambient temperature during the testing sessions was 22–24°C with a relative humidity of 50–55%.

Maximal sprinting speed (1st testing session)

To assess MSS, participants performed 3×40 m-all out sprints with 5 min of passive recovery after a warm up that consisted of 5 min of jogging, 5 min of dynamic stretches and 3×10 m sprints with 30 s recovery. During the 40 m sprints, 10 m split times were recorded using a dual beam photocell timing gate (WinLaborat, Buenos Aires, Argentina). MSS was considered as the fastest speed in one of the 10 m splits across the three repetitions.

RSA test (2nd testing session)

RSA performance was assessed using the protocol proposed by Rampinini et al. (2007). The test consisted of 6×40 m shuttle sprint (20 m + 20 m) with a 20 s passive recovery between sprints. Sprint time was measured using a dual beam photocell timing gate (WinLaborat, Buenos Aires, Argentina). At the end of the test, participants were instructed to remain in seated position for 5 min for HRR data collection. The RSA performance indices calculated were: best sprint time (i.e., the fastest sprint), mean sprint time, total sprint time, ideal sprint time and the percentage of decrement (Sdec), as previously suggested (Dawson, 2012; Girard, Mendez-Villanueva, & Bishop, 2011).

Université of Montréal track test (3rd testing session)

The UMTT was carried out on a 400 m synthetic track as per (Leger & Boucher, 1980) but with the pacing set using a bicycle equipped with a tachometer (Boullosa & Tuimil, 2009). The test started with a running speed of $8 \text{ km} \cdot \text{h}^{-1}$ and increased in $1 \text{ km} \cdot \text{h}^{-1}$ every 2 min. Participants were instructed to run behind the bicycle and to avoid drafting. The test was terminated when the participants could no longer follow the pacing and distanced ≈ 3 m from the bicycle. Maximum aerobic speed (MAS) was defined as the speed achieved in the last completed 2 min stage (Leger &

Boucher, 1980). Upon exhaustion occurrence, participants were instructed to stop and sit down on the ground for 5 min for HRR data collection.

Anaerobic speed reserve

ASR was defined as the difference between the MSS and the MAS, determined during the UMTT. Based on the median value of ASR for males and females, respectively, the sample was divided into two groups: high ASR (males, $n = 13$; females, $n = 7$) and low ASR (males, $n = 14$; females, $n = 7$).

HR data collection

Before the RSA and UMTT tests, participants were equipped with a telemetric HR monitor (Polar RS800 CX, Polar Electro Oy, Finland). HR was then continuously registered beat by beat throughout the tests and during the 5-min post-test recovery. Raw HR data were automatically filtered using a moderate filter (Nunan et al., 2009) and all irregular heartbeats were replaced with interpolated adjacent R-R interval values using the Polar Software (Pro Trainer 5, Polar Electro, Finland). Maximum heart rate (HRmax) was defined as the highest value of HR registered during the tests and HR at the end of tests (HREnd) was defined as the HR registered in the exact moment at which the test ended. Also, HRmax as the percentage of the predicted HRmax (% of predicted HRmax) (Tanaka, Monahan, & Seals, 2001) was calculated. HRR assessment included both raw and relative parameters as well as the HRR kinetics. Raw HR was defined as the HR value at a given point of recovery period (i.e., 20 s, 30 s, 1 min, 2 min, 3 min and 5 min) and relative HR recovery was defined as the difference between HR registered at the end of exercise (HREnd) and after 20 s, 30 s, 1 min, 2 min, 3 min and 5 min (i.e., $\Delta 20''$, $\Delta 30''$, $\Delta 1'$, $\Delta 2'$, $\Delta 3'$ and $\Delta 5'$). For the assessment of HRR, kinetics data were modelled with a monoexponential fit (Sigmaplot 12; SPSS Science, Chicago, IL) as previously described (Boullosa, Barros, Del Rosso, Nakamura, & Leicht, 2014). From this model, the time constant (τ) of the fit (e.g., $HR\tau$ = time to reach a 63% reduction in HR in seconds) and the amplitude of HR decrement of the fit (HRamp) were used for further analysis.

Statistical analyses

Descriptive statistics were used to compute means and standard deviations (SD). The normal distribution of the variables was assessed for all groups in each condition using the Shapiro–Wilk test and visual inspection of Q–Q plots. For those variables showing a non-normal distribution, a natural logarithmic transform was applied while those variables showing a non-normal distribution in spite of transformation were assessed using non-parametric techniques (Mann–Whitney U Test for pairwise comparisons). Comparisons between ASR-based groups and sex groups were performed using a two-way ANOVA (ASR $\times$ sex). Homogeneity of variance was checked using the Levene's test. In case of a significant interaction between groups, a *post hoc* analysis with Bonferroni's comparisons was carried out to assess simple effects. Also, the main effect for each factor was assessed. Pairwise comparisons of HR responses between tests were performed using the t

test for independent samples. Cohen's d was also calculated for assessing the effect size and qualitative statistics were used to assess for chances of the true effects between groups (Hopkins, Marshall, Batterham, & Hanin, 2009) with the following thresholds: most unlikely, $<0.5\%$; very unlikely, 0.5 – 5% ; unlikely, 5 – 25% ; possibly, 25 – 75% ; likely, 75 – 95% ; very likely, 95 – 99.5% and most likely, $>99.5\%$. Qualitative analysis was performed using an available spreadsheet (Hopkins, 2007). All statistics were performed using the IBM SPSS Statistics for Windows® (Version 20.0; Armonk, NY). The statistical significance was set at an alpha level of 0.05.

Results

Indices of RSA and UMTT performance for sex and ASR groups are shown in Table 1. There were no significant sex $\times$ ASR-based group interactions for the test performance indices. Analysis of the main effects revealed that MSS was significantly higher in the high ASR group only in males. In addition, there were significant differences ($P < 0.05$) in performance indices for RSA and UMTT between sex of the same group, except for the Sdec. Also, comparison of the total group mean (males and females collapsed) showed significant differences ($P < 0.05$) between high ASR and low ASR groups for ASR and MSS.

HR and HRR responses are shown in Table 2. There were significant interactions between sex and ASR for HREnd, HR 20'', 30'', 1', 2', and for $\Delta 20''$ and $\Delta 30''$. Also, significant differences were found between high ASR and low ASR groups of the same sex for HRmax, HREnd and indices of relative HRR after the RSA and UMTT. Of note, there was a significant interaction ($F_{1,30} = 10.21$, $P = 0.03$) for $HR\tau$ after the RSA test. The *post hoc* analysis revealed that males from the high ASR group have a significantly slower $HR\tau$ compared with the males of the low ASR ($P < 0.05$) and the females of high ASR ($P < 0.01$), whereas females of the high ASR groups had a

Table 1. Performance indices for the repeated sprint ability test and Université of Montreal Track Test for sex and anaerobic speed reserve groups.

	Sex	High ASR		Low ASR	
		Total mean		Total mean	
ASR ($m \cdot s^{-1}$)	M	4.3 (0.4) ^{a,c}	3.9 (0.7) ^b	3.0 (0.5) ^c	2.7 (0.6)
	F	3.1 (0.4) ^a		2.3 (0.5)	
MAS ($m \cdot s^{-1}$)	M	4.2 (0.4) ^c	3.9 (0.6)	4.2 (0.6) ^c	4.0 (0.6)
	F	3.3 (0.6)		3.5 (0.4)	
MSS ($m \cdot s^{-1}$)	M	8.6 (0.3) ^{a,c}	7.9 (1.1) ^b	7.2 (0.8) ^c	6.8 (1.0)
	F	6.4 (0.4)		5.9 (0.5)	
Best Sprint Time (s)	M	7.8 (0.4) ^c	8.3 (0.8)	8.1 (0.5) ^c	8.5 (0.8)
	F	9.2 (0.6)		9.3 (0.4)	
Mean Sprint Time (s)	M	8.2 (0.3) ^c	8.7 (0.9)	8.5 (0.7) ^c	9.0 (0.9)
	F	9.7 (0.4)		9.9 (0.4)	
Total Sprint Time (s)	M	56.9 (2.1) ^c	60.7 (6.1)	58.8 (4.5) ^c	62.6 (6.3)
	F	65.1 (2.3)		68.0 (3.6)	
Ideal Sprint Time (s)	M	46.6 (2.2) ^c	49.5 (4.9)	48.5 (3.3) ^c	51.2 (4.7)
	F	55.2 (3.4)		56.1 (1.9)	
SDec (%)	M	22.1 (2.6)	22.5 (2.5)	21.1 (2.8)	22.0 (3.5)
	F	19.6 (1.0)		23.3 (3.6)	

Values are means ($\pm$ SD). ^aSignificantly different from Low ASR within the same sex. ^bSignificantly different from Low ASR (males + females collapsed) ($P < 0.05$). ^cSignificantly different between sex within the same group ($P < 0.05$). Total mean = Males and females collapsed. ASR = anaerobic speed reserve; MAS = maximum aerobic speed; MSS = maximal sprint speed; SDec = percentage of decrement in RSA; M = males; F = females.

Table 2. Heart rate responses during and after (i.e., recovery) the repeated sprint ability and Université de Montreal Track Test (UMTT).

	Sex	RSA			UMTT		
		High ASR	Low ASR	ES	High ASR	Low ASR	ES
HRmax (bpm)	M	195 (8) ^c	191 (5) ^c	0.60	209 (6)	204 (6)	0.83
	F	188 (11) ^c	196 (8) ^c	0.83	202 (9)	211 (14)	0.76
% Predicted HRmax	M	101.7 (3.7)	99.7 (3.8)	0.53	109.2 (2.8)	106.4 (3.4)	0.90
	F	98.3 (6.6)	102.3 (4.5)	0.71	105.5 (5.1)	110.2 (7.5)	0.73
HRend (bpm)	M	189 (7) ^c	185 (7) ^c	0.57	202 (7)	201 (6)	0.15
	F	182 (9) ^{a,c}	191 (9) ^c	1.00	198 (8)	206 (12)	0.78
HR 20'' (bpm)	M	179 (9) ^{b,c}	175 (9) ^c	0.44	189 (10)	188 (6)	0.12
	F	168 (15) ^a	185 (10)	1.34	181 (12)	192 (13)	0.88
HR 30'' (bpm)	M	174 (9) ^b	168 (9) ^c	0.67	181 (11)	179 (12)	0.17
	F	158 (17) ^a	179 (9)	1.54	175 (13)	184 (14)	0.66
HR 1' (bpm)	M	157 (10) ^b	151 (12)	0.54	156 (11)	156 (15)	0.00
	F	140 (20) ^a	159 (11)	1.18	152 (10)	160 (22)	0.47
HR 2' (bpm)	M	136 (12) ^b	129 (10)	0.63	129 (12)	132 (9)	0.28
	F	122 (19) ^a	138 (13)	0.98	125 (13)	133 (18)	0.51
HR 3' (bpm)	M	126 (10) ^b	118 (10)	0.80	119 (12)	121 (8)	0.20
	F	111 (24)	126 (11)	0.80	115 (15)	127 (20)	0.68
HR 5' (bpm)	M	119 (9)	114 (10)	0.52	115 (12)	116 (9)	0.09
	F	111 (12)	118 (14)	0.54	113 (16)	122 (16)	0.56
$\Delta 20''^d$	M	10 (5) ^c	10 (4)	0.00	14 (8)	13 (5)	0.15
	F	14 (6) ^a	7 (4) ^c	1.37	17 (5)	14 (8)	0.45
$\Delta 30''^d$	M	15 (5) ^c	17 (6)	0.36	21 (9)	23 (11)	0.20
	F	24 (10) ^a	13 (3) ^c	1.49	23 (7)	22 (6)	0.15
$\Delta 1'^d$	M	31 (6) ^{b,c}	35 (9)	0.52	46 (11)	46 (16)	0.00
	F	42 (13)	32 (4)	1.04	46 (7)	46 (16)	0.00
$\Delta 2'^d$	M	53 (9) ^c	57 (7) ^c	0.49	74 (12)	70 (10)	0.36
	F	60 (12) ^c	53 (10) ^c	0.63	73 (8)	73 (13)	0.00
$\Delta 3'^d$	M	63 (7) ^c	67 (8) ^c	0.53	84 (11)	80 (9)	0.40
	F	71 (18)	66 (10)	0.34	82 (10)	79 (17)	0.21
$\Delta 5'^d$	M	69 (6) ^c	71 (8) ^c	0.28	88 (11)	86 (9)	0.20
	F	70 (7) ^c	73 (13)	0.29	85 (11)	84 (12)	0.08
HRamp (bpm) ^d	M	80.5 (7.3) ^c	82.9 (10.4) ^c	0.27	102.3 (15.2)	90.4 (27.1)	0.54
	F	79.9 (11.2) ^c	83.9 (17.1)	0.27	96.1 (11.7)	94.9 (20.0)	0.07
HR τ (s)	M	104.8 (22.9) ^{a,b,c}	87.5 (15.5) ^c	0.88	74.9 (17.4)	86.5 (17.9)	0.66
	F	71.2 (19.3) ^a	95.8 (13.0)	1.68	73.8 (10.3)	71.2 (10.5)	0.25

Values are means ($\pm$ SD). ^aSignificantly different from Low ASR ($P < 0.05$). ^bSignificantly different from females ($P < 0.05$). ^cSignificantly different from UMTT ($P < 0.05$). ES = effect size; HRmax = maximum heart rate; % Predicted HRmax = HRmax expressed as a percentage of the estimated maximum HR; HRend = Heart rate at the end of the test; HR 20'' – 5' = HR at 20 s, 30 s, 1 min, 2 min, 3 min and 5 min post-test; $\Delta 20' - 5'$ = Difference between HRend and HR at 30 s, 1 min, 2 min, 3 min and 5 min post-test, respectively; HRamp = amplitude of HRR kinetics response; HR τ = time constant for the HR recovery kinetics. ASR = anaerobic speed reserve. ^dAnalyses were performed using a natural logarithmic transformation, but raw data are presented for clarity purposes.

faster HR τ compared with the females of low ASR group. The magnitude-based inferences showed that differences for post-RSA HR τ were likely negative for males, and likely positive for females (Figure 1). When comparing high ASR and low ASR groups collapsed by sex, significant correlations were found between HR τ post-RSA and ASR in the HASR group ($r = 0.66$, $P = 0.04$) and between HR τ post-UMTT and MAS ($r = 0.52$, $P = 0.03$) in the low ASR group. In addition, there was a significant correlation between HR τ post-RSA and ASR ($r = 0.50$, $P = 0.02$), between HR τ and MSS ($r = 0.55$, $P < 0.01$) for males, and a significant inverse correlation ($r = -0.60$, $P = 0.03$) between HR τ post-RSA and ASR in the females.

Discussion

To our knowledge, this is the first study to address whether the ASR could influence post-exercise autonomic responses, as inferred from HRR. The most interesting finding is the significant differences in HR kinetics (i.e., HR τ) between high ASR and low ASR both in males and females; although contrary to our hypothesis, the direction of the difference between the females groups was opposite to that observed between the males groups (Table 2, Figure 1). Interestingly, in comparison

with UMTT, the males of the high ASR group showed a slower HR kinetics post-RSA test despite HRmax on the RSA was being significantly lower than the HRmax registered during the UMTT. Moreover, comparisons of both groups, irrespective of sex, yielded a significant correlation between HR τ post-RSA and ASR in the high ASR group ($r = 0.66$) and between HR τ post-UMTT and MAS ($r = 0.52$) in the low ASR group. Recently, it has been suggested that ARS could influence performance during HIIT (Buchheit & Laursen, 2013) and previously Buchheit et al. (2012) found that ARS (adjusted for MAS) had a very large weight in determining the number of HIIT sets before exhaustion. In this sense, our findings add to the previous reports suggesting that subjects with different ASR could display differences in their HRR. Of note is that we did not adjust ASR for MAS because in our sample the high ASR and low ASR groups had similar values of MAS and the main determinant of ASR was the MSS.

HRR in healthy individuals is governed by the ANS and it is characterised by parasympathetic reactivation and sympathetic withdrawal (Borresen & Lambert, 2008; Daanen et al., 2012). Moreover, since it has been suggested that as ANS interacts with other physiological systems, monitoring of the HRR as a reflection of the ANS in response to an exercise stimulus may reflect body's ability to respond to the stress of

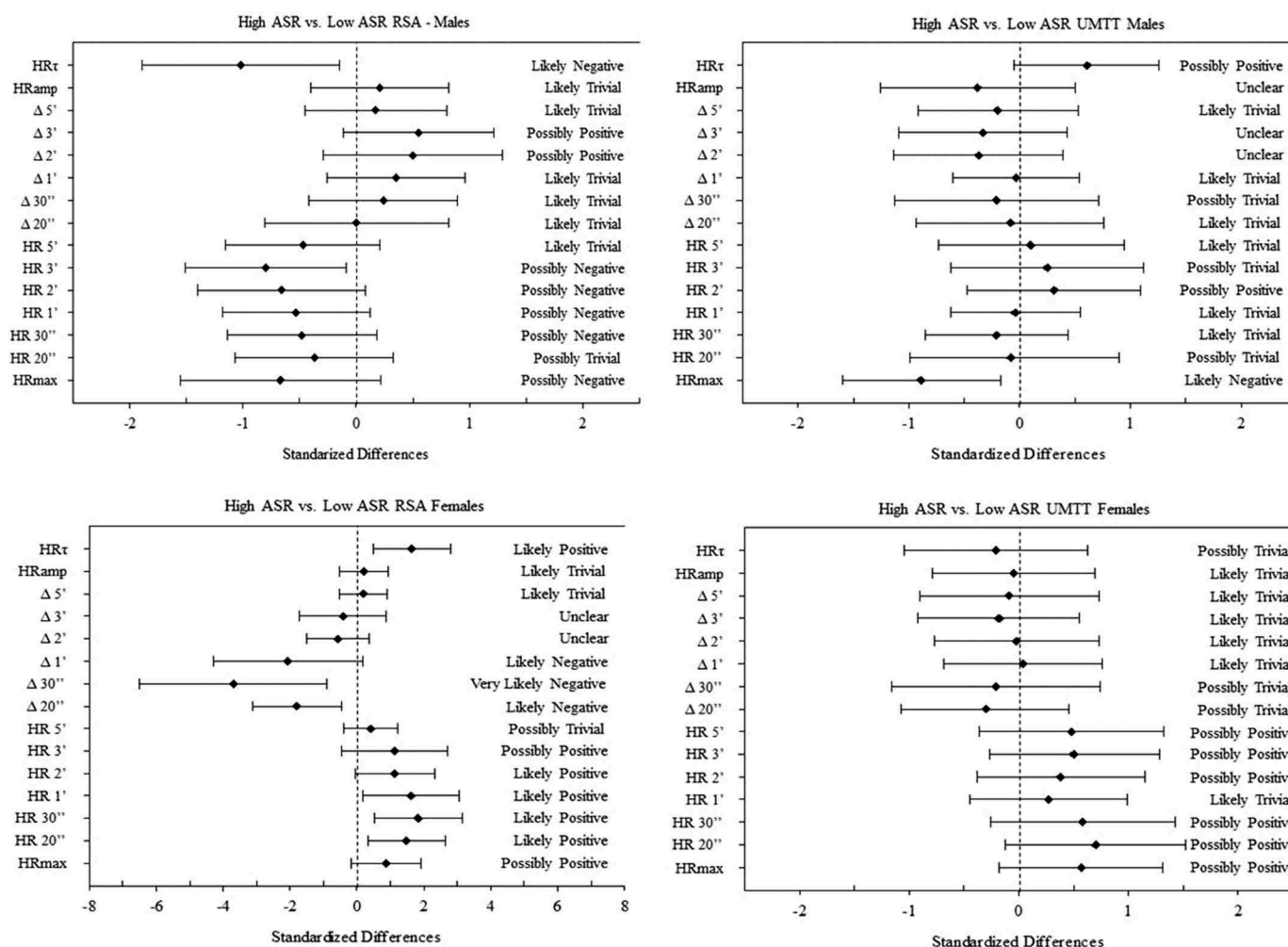


Figure 1. Magnitude-based inferences (standardised differences in Cohen's units) for heart rate and heart rate recovery indices post-RSA (left panels) and UMTT tests (right panels) and for males (upper panels) and females (lower panels) of the high ASR and low ASR groups. HRmax = maximum heart rate; HR 20'' – 5' = HR at 20 s, 30 s, 1 min, 2 min, 3 min and 5 min post-test; $\Delta 20' - 5'$ = Difference between HRend and HR at 20 s, 30 s, 1 min, 2 min, 3 min and 5 min post-test, respectively; HRamp = amplitude of HRR kinetics response; HRt = time constant for the HR recovery kinetics. ASR = anaerobic speed reserve.

such stimulus (Borresen & Lambert, 2007). Our finding of a differential response of HRR kinetics in males and females with different ASR and after different modes of exercises appears to support this contention. Indeed, Buchheit et al. (2007) observed a significantly slower HRR after a repeated-sprint exercise (18 × 15 m sprints with 17 s recovery) in comparison with an isocaloric continuous exercise and a high-intensity intermittent exercise. The authors attributed the impaired parasympathetic reactivation after the repeated-sprint exercise to a high anaerobic demand, a high sympathetic activity and the elevation of adrenergic factors and local metabolites during exercise. Although we did not collect metabolic data, the metabolic responses to a repeated-sprint test and maximal continuous exercises (i.e., UMTT) are well characterised (Edg, Bishop, Hill-Haas, Dawson, & Goodman, 2006; Hamilton, Nevill, Brooks, & Williams, 1991; Lacour, Padilla-Magunacelaya, Chatard, Arsac, & Barthelemy, 1991; Leger & Boucher, 1980; Spencer, Bishop, Dawson, & Goodman, 2005; Spencer, Bishop, Dawson, Goodman, & Duffield, 2006). For instance, in relation with one-set RSA tests, it has been noted that anaerobic power sources are the major determinants of the RSA scores (Dawson, 2012). Moreover, Hamilton et al. (1991) found strong

correlations between peak blood lactate and peak speed ($r = 0.90$, $P < 0.01$) and between peak blood lactate and peak power fatigue ($r = 0.92$, $P < 0.01$) during 10 × 6-s all-out sprints with 30-s recovery periods implying a high glycolytic rate during this activity which has been confirmed by the high lactate levels observed in studies using similar protocols (Dawson, Fitzsimons, & Ward, 1993; Edge, Bishop, Goodman, & Dawson, 2005; Hamilton et al., 1991). In addition, when comparing a continuous (i.e., UMTT) versus a shuttle intermittent test (i.e., 30–15 Intermittent Fitness Test) Buchheit et al. (2009) observed higher values of ventilatory parameters during the 30–15 Intermittent Fitness Test suggesting that metabolic acidosis, and therefore a greater anaerobic contribution, may have contributed to an enhancement of peripheral chemoreceptor activity and resulting in increased ventilation. On the contrary, the UMTT has been designed as a means of estimating $\dot{V}O_{2\max}$ (Leger & Boucher, 1980), thus reflecting the maximal aerobic capabilities in trained and untrained individuals. Therefore, it is possible to suggest that the slower HRR kinetics observed in our male high ASR group could be related to a higher anaerobic contribution and muscular activity during the RSA performance. In addition, the males of the high ASR

group displayed a significantly slower HR_t post-RSA compared to HR_t post-UMTT despite of the significantly higher HR_{max} registered during the UMTT (Table 2). This difference in the HR_t between tests could be related to a higher disturbance in the physiological *milieu* after the RSA test. As a matter of fact, Buchheit, Al Haddad, Laursen, and Ahmaidi (2009) observed an impaired parasympathetic function (e.g., longer HR_t) after a shuttle intermittent test compared with a graded continuous test, which the authors attributed to more disturbed post-exercise homeostasis. Further research on this topic should include metabolic measures to verify whether differences in HRR, in relation with ASR, are associated with the anaerobic contribution or muscle power developed in the exercise.

Surprisingly and contrary to our hypothesis, the female high ASR group showed a faster HRR kinetics compared to both male high ASR and female low ASR groups (Table 2). This is an striking result as previous reports showed an attenuated HRR and parasympathetic reactivation in females versus males after a Wingate test (Barak, Klansja, Gacs, Ovcin, & Grujik, 2014; Mendonca et al., 2010). However, a Wingate test and RSA test may require different anaerobic contributions and therefore may impose differing physiological stress (Meckel, Machnai, & Eliakim, 2009). Also, it appears that posture could have a significant impact on the magnitude of post-exercise HR measures (Buchheit, Buchheit et al., 2009). Thus, comparisons between our results and those from Barak et al. (2014) and Mendonca et al. (2010) may not be appropriate. As we mentioned earlier, a plausible explanation for the difference between males and females in the high ASR group could be a higher anaerobic contribution during the RSA test in the high ASR males compared to the females, mostly due to their ability to develop more absolute speed and power during sprints (Keane, Salicki, Goodall, Thomas, & Howatson, 2015). Accordingly, the males of the high ASR group showed a significantly higher MSS compared to the rest of the groups, whereas the females of high ASR and low ASR groups exhibited similar values of MSS (Table 1). Thus, the higher anaerobic contribution in the high ASR males' group during the RSA test could be accounting for the differential HRR kinetics compared to the females in the high ASR group and the males in the low ASR group.

On the contrary, the differences between high ASR and low ASR groups in females are more difficult to explain. One possibility is that at least half of the females in the high ASR group had a history of intermittent sport practice which was not the case in the females of the low ASR group. In this regard, it has been proposed that ultra-short-term HRR after maximal exercise is faster in athletes of different training background (i.e., intermittent sports vs. continuous sports) (Ostojic et al., 2010). In our study, the females of the high ASR group showed significant higher values of the short-term HRR expressed both in absolute and relative (i.e., $\Delta 20''$ and $\Delta 30''$, Table 2) terms compared with the females of the low ASR group, which is in line with the findings of Ostojic et al. (2010) and could have contributed to the faster HRR kinetics observed. Furthermore, although not significant, the females of the high ASR group showed better sprint and RSA performances than females of the low ASR group and this could be

related to the intermittent training background in some of the females of the high ASR group (Edg et al., 2006). Consequently, the faster HRR kinetics in females of the high ASR group could reflect the specific exercise training history of some of the participants. Moreover, since ASR is a function of MAS and MSS, changes in one of these parameters with training could also affect the relationship between HRR and ASR. For instance, a recent study (McDonald, Grote, & Shoepe, 2014) found significant differences in post-exercise HRR between track and road cyclist which was attributed, in part, to the training background (i.e., aerobically vs. anaerobically trained). Therefore, differences in training programmes for athletes of different sports (e.g., endurance vs. team sports) could lead to different HRR responses. Future longitudinal studies should be conducted for determining the strength and direction of the influence of ASR on HRR in different populations.

Another possibility is that the difference in HR_t between the female groups could be a reflection of differences in physical fitness. It is known that that post-exercise HRR is a marker of training-induced changes in autonomic control (Borresen & Lambert, 2008) and therefore differences in HRR would be related to differences in physical fitness, particularly aerobic fitness which has been demonstrated to influence importantly post-exercise cardiac reactivity (Huang, Webb, Zourdos, & Acevedo, 2013). In our study, although the differences in performance during the tests between the high ASR and low ASR groups were non-significant, when we pooled the data of males and females in one group and calculated quartiles for the tests performances (data not presented), most of the females of the low ASR group were in the 1st quartile in nearly all performance indices, reflecting a lower physical fitness compared with the other groups. Hence, based on our data, we cannot draw a convincing conclusion regarding the role of the ASR and its influence on HRR kinetics for females, and therefore we may suggest that the results of our female groups should be taken with caution. Accordingly, further studies looking at differences in HRR after different modes of exercises in males and females with different metabolic characteristics are warranted. As our participants were physical education students, future studies should look at this in other populations such as trained athletes from different sports, with special attention to the possible influence of the moment of the season (i.e., training status) on these relationships (Aubry et al., 2015; Boullosa, Tuimil, Leicht, & Crespo-Salgado, 2009).

Finally, it is worth mentioning that for this study we have collected the HRR data with the subjects in seated position (i.e., passive recovery) given that previous reports (Bosquet, Gamelin, & Berthoin, 2008) have shown a good reliability of this measure. However, we have recently shown that both HRR and HRR kinetics are highly reliable when measured during walking after a maximal test (Boullosa et al., 2014) which would be a preferable mode of recovery after exercise. Also, given that we found differences between low and high ASR groups, particularly in the HRR kinetics (i.e., HR_t), it would be suggested that that this measure is more sensitive for the evaluation of the post-exercise autonomic control than the other HRR measures, with further longitudinal studies still

needed including the evaluation of these and other HR measures over different days (Tonello et al., 2016). This is important given that both performance tests and autonomic indices could exhibit different reliability levels depending upon the specific sport and the characteristics of the monitored group (e.g., standards) (Boullosa & Abreu, 2014).

In conclusion, we found that HRt post-RSA was slower in the males of the high ASR group compared to the low ASR group, and females of the high ASR group had a faster HRR kinetics compared with males of the high ASR group and females of the low ASR group. Also, the HRt post-RSA in the males of the high ASR group was slower than the HRt post-UMTT. The slower HRR kinetics in the males of the high ASR group could be related to a higher anaerobic contribution to the RSA performance, whereas the faster HRR kinetics in the high ASR females group could be related to other factors and therefore this result should be taken with caution. Further research is needed to assess whether the differential HRR kinetics observed here is related to the ASR in other exercise modes and populations.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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